

Master 4 Grid Maker v3

Operator Manual - Standard Grid + Optional Cell 55 Z10-Z20 TPKX

Master 4 turns one whole-degree geographic box into a repeatable 100-cell production system. Version 3 preserves the standard extent manifest and numbered GeoTIFF, adds a final full-box MASTER extent row, and optionally creates a synthetic colored Cell 55 TPKX for ArcGIS Earth teaching, sanity checking, and offline reference.

Quick start

1. Keep Master4_Grid_Maker.py and Master4_Grid_Maker.bat together.
2. Double-click the BAT and enter: west, east, south, north. Example: 82w, 81w, 30n, 31n.
3. The program first creates the extent TXT and numbered grid GeoTIFF in C:\downloads.
4. It then asks whether to create the Cell 55 colored Z10-Z20 TPKX.
5. The demo is optional. The prompt defaults to No. Press Enter or N to skip it.

Prompt:

Create Cell 55 colored zoom-demo TPKX (Z10-Z20)? [y/N]:

Answer Y/Yes only when you want the large synthetic TPKX. The normal Master 4 files are already complete before this question appears.

Standard output - unchanged production foundation

For 82w, 81w, 30n, 31n the program writes:

C:\downloads\Master4_82W_81W_30N_31N_Extents.txt

C:\downloads\Master4_82W_81W_30N_31N_Grid.tif

The TXT contains 100 numbered production rows plus one final MASTER row. Every numbered row carries the exact QGIS-ready EPSG:3857 extent and deterministic production filename.

01 -9028010.7033, -9016878.7543, 3503549.8435, 3516410.3983

[EPSG:3857] W082W081N031N030-001-GHY-Z20.tpkx

MASTER -9128198.2450, -9016878.7543, 3503549.8435, 3632749.1434 [EPSG:3857]

Cell addressing

Each Master 4 is 10 x 10 geographic cells. Use the first decimal digit of absolute latitude as the tens row and the first decimal digit of absolute longitude plus one as the column. Truncate - do not round. Example: 30.56N, 81.34W -> cell 54. The code handles the four hemisphere combinations so the decimal address remains consistent.

GeoTIFF overlay

The 5000 x 5000 EPSG:3857 GeoTIFF is the visual index. Use it to see cells 01-100 over an overview map in QGIS or ArcGIS Earth.

PERMANENT RULE: Reference overlays ON while planning. Reference overlays OFF before production. Leaving the numbered grid visible during raster production can burn it into the finished imagery.

Optional Cell 55 Z10-Z20 TPKX

If you answer Y/Yes, v3 creates a file such as W082W081N031N030-055-ZOOM-DEMO.tpkx. It contains project-created synthetic PNG tiles at every stored level Z10 through Z20. Each zoom uses a different color and large Z-number plus CELL 55.

The project owner generated the current v3 package on Windows and opened it successfully in ArcGIS Earth. Screenshots showed visible stored-level transitions including Z12, Z13, Z14, Z15, Z17 and Z20. The current field-test package was roughly 1 GB. That size is expected for the dense synthetic pyramid and is not a fixed guarantee.

What it is for

- Teaching: make the hidden raster tile pyramid obvious.
- Pre-QGIS sanity check: prove a local TPKX opens and lands in the expected home Master 4 before producing real imagery.
- Offline reference beacon: when online basemap context disappears, opening the local package gives a known geographic anchor and visible zoom-level feedback.

Coarse-zoom footprint

The demo uses the standard global XYZ tile grid. At low zooms one 256 x 256 tile covers much more ground than a 0.1-degree Master 4 cell, so colored tiles can extend beyond the exact Cell 55 boundary. This is expected. Use the GeoTIFF and manifest for exact boundaries; use the synthetic TPKX for geographic and zoom awareness.

Recommended ArcGIS Earth operating flow

1. Keep area-wide Z17 overview TPKX map(s) enabled for broad context.
2. When an area of interest appears, enable the Master 4 numbered GeoTIFF overlay.
3. Identify the required cell or adjacent cells.
4. Read the matching manifest rows for exact extents and deterministic filenames.
5. Optional: open/create the Cell 55 Z10-Z20 demo TPKX as a fast TPKX/geographic sanity check before QGIS.
6. In QGIS, produce only the high-resolution cells actually needed.
7. Create compatible raster MBTiles, convert them to TPKX, and load the finished cells in ArcGIS Earth.

Z17 overview -> grid -> choose cells -> optional TPKX proof -> QGIS -> detail TPKX

QGIS production reminder

Use the exact extent from the selected manifest row and the filename printed on the same line. Once cell selection is complete, turn the reference grid OFF before producing raster imagery.

Offline use

The synthetic Cell 55 file is useful when the user is offline and ArcGIS Earth no longer has familiar online basemap context. It is a reference convenience, not a substitute for an authoritative offline basemap or certified navigation product.

Requirements and troubleshooting

Requirements: Windows, Python 3, Pillow. Install Pillow once with: `py -3 -m pip install Pillow`

- If the program reports an existing demo TPKX, move, rename or delete the old demo only if you intentionally want to regenerate it.
- If you do not want the large demo file, press Enter at the [y/N] prompt. No cleanup is necessary.
- If colored coverage looks larger than Cell 55 at low zoom, remember the coarse XYZ tile footprint behavior described on page 3.
- If exact production geometry is in question, trust the generated manifest and GeoTIFF boundaries, not the visible outer edge of a low-zoom color tile.

Current v3 identities

`Master4_Grid_Maker.py`

SHA-256 16a42784005743728837d3dfdc2f3295b30071d5ed37ccbdbb9b9ee23e58888

`Master4_Grid_Maker.bat`

SHA-256 abb4e7dd1f1b665e1cf1b9ee4d1bfffdef40e336acb746460640be0a043c20c2d

`Master4_Grid_Maker.zip`

SHA-256 408ba3e129f912b3b4ef0a7df0d6ccc478ff8935ffec9cb235232505e872e158

Scope

The optional color-demo TPKX contains synthetic project-created tiles and downloads no commercial imagery. Real production imagery remains subject to its own permissions and attribution requirements. The Master 4 utility is a practical map-production aid, not a certified navigation or life-safety system.